LESSON 12 · MISSION GUIDE

The Deep Archive

JOINs, subqueries and CASE: questions across tables, and the Vault straight into pandas.

From **Queen Aldara of Thornmere**
To **Jen, Keeper of the Guild Ledger**
By **raven, at dusk on the last day of the week**

Keeper, Orrin tells me the Vault has a lower floor, the Deep Archive, where the tables are kept apart: heroes on one shelf, quests on another, who-went-where on a third. The answers I need for the Charter sit between the shelves. Which kinds of hero take the hard quests. Which regions we pay for and can't even find on the map.

He says the word for fetching from several shelves at once is **JOIN**, and that one of the shelves has a hole in it. I would like the answers, and I would like the hole found before the Royal Auditors find it. No pressure. (Some pressure.) — A.

What you'll be able to do after today

- [ ] **JOIN** two and three tables with `ON`, using short table aliases
- [ ] Keep unmatched rows with **LEFT JOIN**, find them with `IS NULL`, fill them with `COALESCE`
- [ ] Build categories and conditional counts with **CASE WHEN**
- [ ] Ask a question inside a question: **subqueries**, plain and per row
- [ ] Name a step with **WITH**, and pass values safely with `?`
- [ ] Pull a query straight into a DataFrame with `pd.read_sql_query`

Why this matters

- **Real databases are many tables.** Nearly every useful question needs a JOIN. This is the lesson interviewers test.
- **SQL + pandas is the everyday kit.** SQL fetches exactly the rows you need; pandas shapes them into the report.

Four big ideas

1. JOIN = XLOOKUP, all rows	`FROM party AS p JOIN heroes AS h ON p.hero_id = h.hero_id`. Inner keeps only matches.
2. LEFT JOIN keeps the left	Unmatched rows get NULL on the right. `IS NULL` finds them, `COALESCE` fills them.
3. CASE is IF	`CASE WHEN reward >= 1000 THEN 'Rich' ELSE 'Small' END`. Inside SUM it counts conditionally.
4. Queries nest	A subquery in brackets, or a named step with `WITH`. Values from Python go in with `?`.

Three things that will bite you

- A JOIN without `ON` pairs every row with every row. A column in two tables needs its alias: `p.quest_id`.
- Joining a one-row-per-hero table repeats each quest's numbers once per hero. Sum with care.
- `= NULL` is never true. And never glue values into SQL with an f-string: use `?`.

How today works

15 MIN
Lecture
Lars walks the Deep Archive, shelf to shelf, and finds the hole

60 MIN
Exercise sheet
Seven ravens, from first JOIN to pandas

15 MIN
Pages from the Grimoire, page 12
"deep_archive.py", round a candle stub

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Siv	The first JOIN	`JOIN … ON`, aliases
2	The Queen	Three tables	two JOINs, `GROUP BY`
3	Orrin	The missing region	`LEFT JOIN`, `IS NULL`, `COALESCE`
4	Pell	CASE WHEN	`CASE`, `SUM(CASE …)`
5	Siv	Questions inside questions	subqueries, `NOT IN`
6	Orrin	WITH, and asking with ?	`WITH`, `?` parameters
7 ⭐	Pell	Into pandas	`pd.read_sql_query`, crosstab, pivot_table

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if Pell gets her tables), each printing the expected output
- [ ] Dalvric's deep_archive shows all five regions and the right crowns, one commit per bug
- [ ] Everything is pushed to `lesson-12` and you've opened a pull request



Next lesson: Before the Throne. The numbers are found. Now they need to look like something a queen will read: charts with matplotlib, and a properly formatted Excel workbook written straight from Python.